



**Hubble revealed** that galaxies are moving away from Earth with speeds proportional to their distance. The result of this motion is that as time passes, the galaxies themselves are also moving farther apart, meaning that the Universe is expanding.

“The **history of astronomy** is a history of **receding horizons.**”

Edwin Hubble, 1936

GEORGES LEMAITRE

**Belgian astronomer, cosmologist, and Roman Catholic priest Georges Lemaître was the first scientist to propose the theories of an expanding Universe and the Big Bang.**



In 1927, Lemaître (1894–1966) published a paper hypothesizing that the Universe is expanding. While his theory was largely overlooked at the time, today scientists agree that Lemaître should share credit for the discovery with Hubble, whose findings were published 2 years later. In 1931, Lemaître proposed that the Universe began as a single dense particle from which all energy and matter derived. This idea formed the basis of the Big Bang theory, evidence for which was discovered in 1964.

Hubble used the Hooker telescope to observe Cepheid variable stars (stars whose luminosity pulsates) within the spiral nebula Andromeda. By plotting the Cepheids' periods (the time for one cycle) of variation, he established the stars' absolute magnitude (full luminosity) and then compared this with their apparent magnitude (brightness from Earth) to calculate how far away they were. The results revealed that Andromeda is a vast, independent galaxy that is comparable in size to the Milky Way and located far beyond it—almost